SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	S.B. Sayed Basha	**Reg. No.:**	192372357
Section / Batch:	CSE(AI)/2023	**Date:**	9/10/2026

Code of Conduct

I S.B. Sayed Basha (192372357) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Sayed Basha

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Dr. J Gokulnath		
Signature:	Signature:	Signature:
Date: 09/10/2026	Date:	Date:

42/100

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CO-1 AT2

Test - Mobile Computing

S.B. Sayed Basha
19237357
ITA0301
Mobile Computing

1) Telecom operator wins a spectrum auction = 45 MHz
company policy set aside = 25 KHz

$= 45000\ KHz + 25\ KHz = 45025\ KHz$

$= \frac{45000\ KHz}{25\ KHz}$

channels = 1800 Hz

2) A regional carrier runs its network = 560 channels
reuse cluster = 7-cell frequency
How many channels for a single cell?

560 channels → 7-cells
T? → 1 cells

$7T = 560$

$T = \frac{560}{7} = 80$ channels

3) GSM operator set carrier = 200 KHz
rollout = 100 carriers
Each carrying = 8 time slots

Total = $200 \times 1000\ Hz * 100 \times 8$
$= 2000000 \times 100 \times 8$
$= 16\ 0000000$ channels
$= 160$ Mchannels.

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5) 80w - 4w = 76w

4) $\frac{-60\ dBm}{-92\ dBm} = 0.65$ = $P_1 - P_2$ = $-60 - (-92)$ = 32 dBm.

6) $\frac{42\ MHz}{200\ KHz} = \frac{42000\ KHz}{200\ KHz} \cdot \frac{140}{7}$ = 20 channels per single cell

7) $\frac{64w}{4} = 16$

8) Inverse Square's Power law

9) 15x4 = 60 channel

10) $\frac{900 \times KHz}{100\ KM/H} = \frac{900 \times 1000\ KHz}{100\ KM/H} = 9000$

11) $\frac{5}{8} = 0.62$

12) $\frac{24\ MHz}{200\ KHz} = \frac{24000\ KHz}{200}$ = 120 carrier frequencies

13) 33dB x 8 = 264 dB

14) 1900 x 2.5 = 190 x 25 =

15) 250 x 6 = 1500 clusters

16) $\frac{4000}{3 \times 60} = \frac{400}{18}$ = 22.2 calls = 4000 x 3 x 60 = 720000 calls

17) 150 x 25000 = 3750000

18) 20 x 40 KHz = 1000 K bandwith

19) 1900 x 2.5 = 4750

20) 8 x 40 x 250 = 8 x 10000 = 80000

21) 400 x 400 = 160000 sq km.

$\frac{000\ KH2}{20\ KH2}$. 200 channels

$\frac{1200 \times 4}{2.5} = \frac{4800}{2.5} = \frac{48000}{25} = 1920$

$\frac{4000\ KH2}{4} = 100 \times 2.5 = 250$

$1200 \times 600 = 720000$ user capacity

$800 \times 4 = 800$ paging rounds

$140,000 - 40,000 = 60000 \times 3 = 180,000$

$\frac{1000000}{100} \times 15 = 700000 = 700,000$ subscribers

$\frac{9000}{12000} = 0.75 \times 100\%$

$= 75\%$

$\frac{48\ MH2}{200\ KH2} = \frac{48\ 000\ KH2}{200\ KH2} = \frac{480}{2} = 240 \times 4 = 960$ channels

$\frac{930 \times 1000\ KH2}{200\ KH2} = \frac{9300}{2} = 4650$

$\frac{2500}{100} \times 99 = 2475$ successfully handed off

$45 \times 18 = 540$

$1700 \times 7 = 6800$ clusters

$4.8 \times 8 = 38.4$ MB

$10 \times 18 = 15 \times 18 = 270$

$250,000 - 70,000 = 180,000 \times 4 = 720,000$

$600 \times 9 = 5400$ clusters

39) $\frac{60000}{600} \times \frac{4}{10} = 240$ calls

40) $150 \times 2.5 = 325\ km^2$

40) $437 \div 7 = 180$

41) $150 \times 2.5 = 325\ km^2$

42) $\frac{48\ MHz}{200\ KHz} = \frac{48000\ KHz}{200\ KHz} = 240 \times 6 = 1440$

43) $\frac{150000}{6000} = 25$ paging cycles

44) $\frac{9}{3} = 3$ handoffs

45) $85 : 115$

$85 : 115$

$17 : 23$

Signal-to-noise ratio = $17 : 23$

46) $250 \times 150\ Kbps = 32500\ Kbps = 32.5\ Mb$ bandwidth

47) $1500 \times 4 \times 2.5 = 600 \times 25 = 15000$

48) $250 \times 12 = 3000$ bandwidth

49) $1500 \times 600\ Kbps = 900000 = 900\ Mbps$

50) $250 \times 4 = 1400\ km^2$